MIT ZARAGOZA
International Logistics Program

# MZ530 Financial Systems 2026/2027

# Assignment #3

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- **Individual Assignment**
- **Due: Oct 29th (Thu): 08.30 am**
- **Please submit you work online via Canvas (only one .pdf file named MZ530_HW3_Name).**

## Problem 1 (Max Points 10)

Consider the following three mutually exclusive projects

| Period | A | B | C |
|---|---|---|---|
| 0 | -3,400 | -3,400 | -3,400 |
| 1 | 0 | 1,870 | 1,020 |
| 2 | 0 | 1,870 | 1,700 |
| 3 | 6,460 | 1,870 | 3,284 |

Rank the projects using each of these decision criteria:

- Payback period
- IRR
- NPV with 10% discount rate
- NPV with 15% discount rate

## Problem 2 (Max Points 10)

Given these two mutually exclusive projects:

| | Project | |
|---|---|---|
| Period | A | B |
| 0 | -400 | -200 |
| 1 | 250 | 140 |
| 2 | 300 | 179 |

- Calculate IRR
- If discount rate is 9%, which project should be selected? Please **explain why**.

## Problem 3 (Max Points 40)

ABC Manufacturing is considering an investment in new automated equipment that is expected to increase productivity significantly. The equipment costs $800,000 and is expected to have a useful life of 5 years. The annual cash inflows resulting from increased productivity are estimated at $200,000.

The company's existing assets, which would be replaced by the new equipment, will be sold for $180,000. The remaining useful life of those assets is 3 years, their (remaining) book value is $300,000, and they are being depreciated at an annual rate of $100,000.

The new equipment is also expected to result in cost savings, including a reduction in operating expenses (headcount reduction) by $50,000 net per year. Additionally, to support the new equipment, ABC Manufacturing estimates an initial investment in spare parts and consumables inventory of $30,000, which is expected to be replenished as consumed for the entire duration of the project (5 years).

The company's discount rate is 15%, while tax rate is equal to 30%.

Questions:

1. Calculate the annual cash flows associated with the new equipment.
2. Calculate the Net Present Value (NPV) of the investment and advise ABC Manufacturing on whether to proceed with the project.

## Problem 4 (Max Points 40)

Firm C2L manufactures a car component whose production will be discontinued in 3 years. C2L is considering changing its current production strategy from chase to level for the next three years. A preliminary study (whose cost has been $50K) indicates that, if the new strategy is adopted, the expected relevant changes are:

1. Some old machines will be sold at the beginning of the project (say time 0). Overall purchasing cost was $1,000K, accumulated depreciation up to the beginning of the project is $400K, and remaining life for the machines is 3 years, according to the accounting department. A firm showed interest to pay $300K (cash) for the machines.
2. Average inventory will increase dramatically (by $150K) at the beginning of the project, and then will remain that high for three years. After three years, inventory will be liquidated at acquisition cost.
3. New software will be necessary to control for inventory. The cost of the software is 300K, to be paid at the beginning of the project. It will be (linearly) depreciated in three years.
4. Savings from reducing overtime will be 100K in the first year and are expected to

increase 10% every year.

C2L has had a positive net profit for many years, and net profit is expected to remain positive no matter what decision is made. Tax rate is 40%. The discount rate of the firm is 20%.

**Question 1**: Calculate the **NPV for change #1 only** if a level strategy is adopted (15 points)

**Question 2**: Calculate the total non-discounted relevant cash flows at the end of years 0, 1, 2, 3. Show your work. Ignore inventory holding cost (insurance, handling …). Your final answer should look like this: (20 points)

| | Time 0 | End of Year 1 | End of Year 2 | End of Year 3 |
|---|---|---|---|---|
| Relevant flows<br>……. | *Your answer* | *Your answer* | *Your answer* | *Your answer* |

**Question 3**: Using an economic (NPV) criterion, should level production be adopted? (5 points)